COS202 ASSIGNMENT REPORT
Student Information
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Department: COMPUTER SCIENCE
Faculty: COMPUTING
Course Code: COS202
Course Title: COMPUTER PROGRAMMING II

ASSIGNMENT 1
TITLE
Development of a Mathematical Calculator (MC) Using Python

OBJECTIVE
The objective of this project is to develop a simple Mathematical Calculator using Python programming language. The calculator performs basic arithmetic operations using selection control statements (if, elif, and else) and allows users to repeatedly perform calculations until they choose to exit.

PROGRAMMING LANGUAGE
Python

DEVELOPMENT ENVIRONMENT
Pydroid 3

PROGRAM DESCRIPTION
The program begins by displaying "MATHEMATICAL CALCULATOR" on the screen.
The user is then prompted to choose one of the available operations:
 *Addition (+)
 *Subtraction (-)
 *Multiplication (*)
 *Division (/)
 *Floor Division (\)
 *Modulus (%)
 *Exponent (^)
 *Clear (C)
 *OFF (Exit)
For every arithmetic operation, the program requests two numbers from the user and performs the selected calculation.
If division, floor division, or modulus is selected and the second number is zero, the program displays "Cannot divide by zero" to prevent an error.
If the user enters C, the program displays "Screen Cleared".
If the user enters OFF, the calculator terminates.
If an invalid option is entered, the program displays "Invalid Operation".

FEATURES OF THE CALCULATOR

*Performs addition
*Performs subtraction
*Performs multiplication
*Performs division
*Performs floor division
*Performs modulus
*Performs exponentiation
*Handles division by zero
*Clears the screen
*Exits the program

SCREENSHOT OF PROGRAM EXECUTION

```
MATHEMATICAL CALCULATOR

Enter (+, -, *, /, \, %, ^, C, OFF): +
First Number: 5
Second Number: 4
Answer = 9.0

Enter (+, -, *, /, \, %, ^, C, OFF): -
First Number: 5
Second Number: 4
Answer = 1.0

Enter (+, -, *, /, \, %, ^, C, OFF): *
First Number: 5
Second Number: 4
Answer = 20.0

Enter (+, -, *, /, \, %, ^, C, OFF): /
First Number: 5
Second Number: 4
Answer = 1.25

Enter (+, -, *, /, \, %, ^, C, OFF): \
First Number: 5
Second Number: 4
Answer = 1.0

Enter (+, -, *, /, \, %, ^, C, OFF): %
First Number: 5
Second Number: 4
Answer = 1.0

Enter (+, -, *, /, \, %, ^, C, OFF): ^
First Number: 5
Second Number: 4
Answer = 625.0

Enter (+, -, *, /, \, %, ^, C, OFF): c
Screen Cleared

Enter (+, -, *, /, \, %, ^, C, OFF): OFF
Calculator Off

[Program finished]
```

CHALLENGES ENCOUNTERED

During the development of this calculator, I encountered some challenges such as:

*Understanding how to use while True to keep the calculator running continuously.

*Preventing division by zero errors.

*Using if and elif statements correctly for different operations.
These challenges were solved through testing and debugging.

CONCLUSION
This project improved my understanding of Python programming, especially the use of loops, user input, arithmetic operators, and selection control statements.

ASSIGNMENT 2
TITLE
Development of a Personal Pocket CGPA Calculator (PPC) Using Python

OBJECTIVE
The objective of this project is to develop a Personal Pocket CGPA Calculator that enables students to calculate their CGPA by entering course units and grades obtained in both the first and second semesters.

Programming Language
Python

Development Environment
Pydroid 3

PROGRAM DESCRIPTION
The program begins by requesting the student's name.
It then asks for the number of courses offered during the first semester.
For each course, the program requests:
*Course Unit
*Grade (A–F)
The entered grades are converted into grade points using selection control statements (if, elif, and else) as follows:
*A = 5
*B = 4
*C = 3
*D = 2
*E = 1
*F = 0
The same procedure is repeated for the second semester.
After collecting all the course information, the program calculates:
Total Course Units (TCU)
Total Grade Points (TGP)
CGPA = Total Grade Points ÷ Total Course Units
The CGPA is rounded to two decimal places and displayed together with the student's name.
The program also determines the student's class of degree based on the calculated CGPA.

Degree Classification
*4.50 – 5.00 → First Class
*3.50 – 4.49 → Second Class Upper
*2.40 – 3.49 → Second Class Lower
*1.50 – 2.39 → Third Class
*1.00 – 1.49 → Pass
*Below 1.00 → Fail

FEATURES OF THE CGPA CALCULATOR
*Accepts student's name.
*Accepts course units and grades.
*Calculates Total Course Units.
*Calculates Total Grade Points.
*Computes CGPA.
*Displays the student's degree classification.

SCREENSHOT OF PROGRAM EXECUTION

```
PERSONAL POCKET CGPA CALCULATOR
Student Name: Anibolu Eze

Number of First Semester Courses: 9

FIRST SEMESTER
Course Unit: 2
Grade (A-F): B
Course Unit: 2
Grade (A-F): B
Course Unit: 2
Grade (A-F): A
Course Unit: 2
Grade (A-F): A
Course Unit: 2
Grade (A-F): B
Course Unit: 2
Grade (A-F): D
Course Unit: 2
Grade (A-F): C
Course Unit: 3
Grade (A-F): B
Course Unit: 2
Grade (A-F): B

Number of Second Semester Courses: 9

SECOND SEMESTER
Course Unit: 2
Grade (A-F): C
Course Unit: 2
Grade (A-F): B
Course Unit: 2
Grade (A-F): B
Course Unit: 2
Grade (A-F): B
Course Unit: 2
Grade (A-F): C
Course Unit: 2
Grade (A-F): B
Course Unit: 3
Grade (A-F): c
Course Unit: 2
Grade (A-F): a
Course Unit: 2
Grade (A-F): b

Student: Anibolu Eze
CGPA = 3.82
Class: Second Class Upper

[Program finished]
```

CHALLENGES ENCOUNTERED
Some challenges encountered during this project include:
*Converting letter grades to grade points.
*Calculating total course units and total grade points accurately.
*Implementing the degree classification using if and elif statements.
*Testing the program with different inputs to verify the correctness of the results.
These challenges were overcome through testing and careful debugging.

CONCLUSION
This project enhanced my understanding of Python programming by applying loops, variables, user input, arithmetic operations, and selection control statements to solve a real-life problem. It also strengthened my problem-solving and programming skills.